

Lognormal Distribution

①

Let 'x' be a positive r.v and let 'y' be a new r.v, defined as $y = \log x$. If 'y' has a Normal distribution, then 'x' is said to be lognormal dist.

The p.d.f. of the lognormal distribution is given by.

$$f(x) = f(x; \mu, \sigma) = \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\log x - \mu)^2}$$

put $\log x = y$
 $\frac{1}{x} dx = dy$
 $0 < x < \infty$
 $-\infty < y < \infty$
 $\sigma > 0$

Assignment

① Question Show that the area under log normal distribution is unity.

② Question Find mean and Variance of the lognormal distribution.

Sol $\mu'_1 = E(x) = \int_0^{\infty} x \cdot \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\log x - \mu)^2} dx$

$$= \int_0^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-\left(\frac{\log x - \mu}{\sqrt{2}\sigma}\right)^2} dx$$

$$\frac{\log x - \mu}{\sqrt{2}\sigma} = y$$

$$x=0, y=-\infty$$

$$x=\infty, y=\infty$$

$$\log x - \mu = \sqrt{2}\sigma y$$

$$\log x = \mu + \sqrt{2}\sigma y$$

$$x = e^{\mu + \sqrt{2}\sigma y}$$

$$dx = e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$\mu' = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{-y^2} e^{\mu + \sqrt{2}\sigma y} \sqrt{2} dy$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2} e^{\mu} e^{\sqrt{2}\sigma y} dy$$

$$= \frac{e^{\mu}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-y^2 + \sqrt{2}\sigma y} dy$$

$$= \frac{e^{\mu}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y^2 - 2\sigma y \frac{\sqrt{2}}{\sqrt{2}} + (\frac{\sigma}{\sqrt{2}})^2 - (\frac{\sigma}{\sqrt{2}})^2)} dy$$

$$= \frac{e^{\mu}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y - \frac{\sigma}{\sqrt{2}})^2 + \frac{\sigma^2}{2}} dy$$

$$= \frac{e^{\mu} e^{\frac{\sigma^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y - \frac{\sigma}{\sqrt{2}})^2} dy$$

$$y = \frac{\sigma}{\sqrt{2}}$$

$$dy = dz$$

$$\begin{array}{c|c} y = -\infty & y = \infty \\ z = -\infty & z = \infty \end{array}$$

$$\frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} dz$$

$$\frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-z^2} dz \quad \begin{array}{l} z' = w \\ dz = dw \end{array}$$

$$\frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_0^{\infty} e^{-w} \cdot \frac{dw}{\sqrt{w}} = \frac{dw}{2\sqrt{w}}$$

$$\frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_0^{\infty} e^{-w} \cdot w^{\frac{1}{2}-1} dw$$

$$\frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \cdot \frac{\sqrt{\pi}}{2} = \frac{e^{\frac{\mu + \sigma^2}{2}}}{2}$$

$$\mu_1' = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{1}{x\sqrt{x}} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

$$= \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} z^{\frac{1}{2}-1} e^{-z} dz$$

$$= \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \cdot \frac{1}{2} = \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}}$$

$$\mu_1' = \frac{\mu + \sigma^2}{2}$$

$$\mu_2' = E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x^2 e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

$$\frac{\log x - \mu}{\sqrt{2}\sigma} = y \quad \begin{array}{l|l} x=0 & x=\infty \\ y=-\infty & y=\infty \end{array}$$

$$\log x = \mu + \sqrt{2}\sigma y$$

$$x = e^{\mu + \sqrt{2}\sigma y}$$

$$dx = e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{\mu + \sqrt{2}\sigma y} \cdot e^{-y^2} \cdot e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{2\mu + 2\sqrt{2}\sigma y - y^2} dy$$

$$\mu_1' = \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{\frac{\mu + \sigma^2}{2}} \cdot e^{-z^2} \cdot \frac{1}{\sqrt{2}} dz$$

$$= \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \int_{-\infty}^{\infty} z^{\frac{1}{2}-1} e^{-z^2} dz$$

$$= \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}} \cdot \sqrt{\frac{1}{2}} = \frac{e^{\frac{\mu + \sigma^2}{2}}}{\sqrt{\pi}}$$

$$\mu_1' = e^{\frac{\mu + \sigma^2}{2}}$$

$$\mu_2' = E(x^2) = \int_{-\infty}^{\infty} x^2 \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(\log x - \mu)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x^2 e^{-\frac{(\log x - \mu)^2}{2\sigma^2}} dx$$

$$\frac{\log x - \mu}{\sqrt{2}\sigma} = y$$

$$x = 0 \quad | \quad x = \infty$$

$$y = -\infty \quad | \quad y = \infty$$

$$\log x = \mu + \sqrt{2}\sigma y$$

$$x = e^{\mu + \sqrt{2}\sigma y}$$

$$dx = e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} e^{\mu + \sqrt{2}\sigma y} \cdot e^{-y^2} \cdot e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{2\mu + 2\sqrt{2}\sigma y - y^2} dy$$

$$\mu_2' = \frac{2\mu}{\sqrt{\pi}} \int_{-\infty}^{\infty} \frac{e^{-\left(y^2 - 2\sqrt{2}\sigma y + (\sqrt{2}\sigma)^2 - (\sqrt{2}\sigma)^2\right)} dy}{e^{\frac{2\mu}{\sqrt{2}\sigma}y - \frac{\mu^2}{\sigma^2}}} \quad (7)$$

$$= \frac{2\mu + 2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y - \sqrt{2}\sigma)^2} dy$$

$y - \sqrt{2}\sigma = z$
 $y - \sqrt{2}\sigma = \sqrt{z}$
 $y = \sqrt{z} + \sqrt{2}\sigma$

put $y - \sqrt{2}\sigma = z$
 $dy = dz$

$$V(X) = \mu_2 - (\mu_1)^2$$

$$= e^{2\mu + 2\sigma^2} - \left(e^{\mu + \frac{\sigma^2}{2}} \right)^2$$

$$= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$$

$$\mu_2 - \mu_1^2 = e^{2\mu + \sigma^2} \left[e^{\sigma^2} - 1 \right]$$

Question Show that in a log normal distribution

$$\mu_x = e^{\mu + \frac{\sigma^2}{2}}$$

Also find β_1 and β_2

Sol By definition.

$$\mu_x' = E(X^r) = \int_0^{\infty} x^r \cdot \frac{1}{x\sqrt{2\pi}\sigma} e^{-\left(\frac{\log x - \mu}{\sqrt{2}\sigma}\right)^2} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_0^{\infty} x^{r-1} e^{-\left(\frac{\log x - \mu}{\sqrt{2}\sigma}\right)^2} dx$$

$$\frac{\log x - \mu}{\sqrt{2}\sigma} = y \quad \begin{matrix} x=0, & y=-\infty \\ x=\infty, & y=\infty \end{matrix}$$

$$\log x = \mu + \sqrt{2}\sigma y$$

$$x = e^{\mu + \sqrt{2}\sigma y}$$

$$dx = e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} \left(e^{\mu + \sqrt{2}\sigma y} \right)^{r-1} e^{-y^2} \cdot e^{\mu + \sqrt{2}\sigma y} \cdot \sqrt{2}\sigma dy$$

$$= \frac{1}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{\mu(r-1) + (r-1)\sqrt{2}\sigma y} \cdot e^{-y^2} \cdot e^{\mu + \sqrt{2}\sigma y} dy$$

$$= \frac{e^{\mu r - \mu}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{\sqrt{2}\sigma y} \cdot e^{-y^2} \cdot e^{\sqrt{2}\sigma y} dy$$

⑥

$$\mu_y' = \frac{\mu_x}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(y^2 - r\sqrt{2}\sigma y)} dy$$

$$= \frac{\mu_x}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(y^2 - 2r \frac{\sigma}{\sqrt{2}} y + \left(\frac{r\sigma}{\sqrt{2}}\right)^2 - \left(\frac{r\sigma}{\sqrt{2}}\right)^2\right)} dy$$

$$= \frac{\mu_x + \frac{r^2 \sigma^2}{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-\left(y - r \frac{\sigma}{\sqrt{2}}\right)^2} dy$$

ملاحظة: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$

$$\frac{\mu_x + \frac{r^2 \sigma^2}{2}}{\sqrt{\pi}} \cdot \sqrt{\pi}$$

$y - r \frac{\sigma}{\sqrt{2}} = z$
 $dy = dz$
 $z^2 = z$

Moments about origin

⑦

$$\mu'_1 = e^{\mu + \frac{1}{2}\sigma^2}$$

$$\mu'_2 = e^{2\mu + 2\sigma^2}$$

$$\mu'_3 = e^{3\mu + \frac{9}{2}\sigma^2}$$

$$\mu'_4 = e^{4\mu + 8\sigma^2}$$

Moments about mean

$$\mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2 = e^{2\mu + 2\sigma^2} - \left(e^{\mu + \frac{1}{2}\sigma^2}\right)^2$$

$$= e^{2\mu + 2\sigma^2} - e^{2\mu + \sigma^2}$$

$$\mu_2 = e^{2\mu + \sigma^2} (e^{\sigma^2} - 1)$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$= e^{3\mu + \frac{9}{2}\sigma^2} - 3(e^{2\mu + 2\sigma^2})(e^{\mu + \frac{1}{2}\sigma^2})$$

$$+ 2(e^{\mu + \frac{1}{2}\sigma^2})^3$$

$$= e^{3\mu + \frac{9}{2}\sigma^2} - 3e^{3\mu + \frac{5}{2}\sigma^2} + 2e^{3\mu + \frac{3}{2}\sigma^2}$$

$$\mu_3 = e^{3\mu} \left(e^{\frac{9}{2}\sigma^2} - 3e^{\frac{5}{2}\sigma^2} + 2e^{\frac{3}{2}\sigma^2} \right)$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

$$= e^{4\mu + 8\sigma^2} - 4(e^{\mu + \frac{1}{2}\sigma^2})(e^{3\mu + \frac{9}{2}\sigma^2}) + 6(e^{2\mu + 2\sigma^2})(e^{\mu + \frac{1}{2}\sigma^2})^2 - 3(e^{\mu + \frac{1}{2}\sigma^2})^4$$

$$\mu_4 = e^{4\mu+8\sigma^2} - 4e^{4\mu+5\sigma^2} + 6e^{4\mu+3\sigma^2} - 3e^{4\mu+2\sigma^2} \quad (8)$$

$$\mu_4 = e^{4\mu} (e^{8\sigma^2} - 4e^{5\sigma^2} + 6e^{3\sigma^2} - 3e^{2\sigma^2})$$

$$\beta_1 = \frac{(\mu_3)^2}{(\mu_2)^3}$$

$$= \frac{\left[e^{3\mu} (e^{\frac{9}{2}\sigma^2} - 3e^{\frac{5}{2}\sigma^2} + 2e^{\frac{3}{2}\sigma^2}) \right]^2}{\left[e^{2\mu+\sigma^2} (e^{\sigma^2} - 1) \right]^3}$$

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2}$$

$$= \frac{e^{4\mu} (e^{8\sigma^2} - 4e^{5\sigma^2} + 6e^{3\sigma^2} - 3e^{2\sigma^2})}{\left[e^{2\mu+\sigma^2} (e^{\sigma^2} - 1) \right]^2}$$

Question If $\log x$ is normally distributed with mean 1 and Variance 4 then find $p(\frac{1}{2} < x < 2) = ?$

Sol

Ans we know that

$$f(x) = \frac{1}{x\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \left(\frac{\log x - \mu}{\sigma} \right)^2}$$

$$\mu = 1, \sigma = 2$$

$$\mu = 1$$

$$\sigma^2 = 4$$

$$\sigma = 2$$

$$f(x) = \frac{1}{2x\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\log x - 1}{2}\right)^2}$$

⑨

$$P\left(\frac{1}{2} < x < 2\right)$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{1/2}^2 \frac{1}{x} e^{-\frac{1}{2}\left(\frac{\log x - 1}{2}\right)^2} dx$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-0.693}^{0.693} \frac{1}{x} e^{-\frac{1}{2}\left(\frac{t-1}{2}\right)^2} x dt$$

$$\log x = t$$

$$\frac{1}{x} dx = dt$$

$$dx = x dt$$

$$x = \frac{1}{2} \quad t = \ln \frac{1}{2} = -0.693$$

$$x = 2 \quad t = \ln 2$$

$$= \frac{1}{2\sqrt{2\pi}} \int_{-0.693}^{0.693} e^{-\frac{1}{2}\left(\frac{t-1}{2}\right)^2} dt$$

$$\frac{t-1}{2} = z$$

$$t-1 = 2z$$

$$t = 1 + 2z$$

$$dt = 2dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-0.8465}^{-0.1535} e^{-\frac{1}{2}z^2} dz$$

$$t = -0.693 \quad z = -0.8465$$

$$t = 0.693 \quad z = -0.1535$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-0.8465}^{-0.1535} e^{-\frac{1}{2}z^2} dz$$

$$P(-0.8465 < z < -0.1535)$$

$$\Phi(-0.1535) - \Phi(-0.8465)$$

$$0.2996 - 0.0596$$

$$\underline{\underline{0.24}}$$